

## Clusteranalyse - Patty Sausele 1146363

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler, MinMaxScaler,
LabelEncoder
import plotly.express as px
import plotly.graph_objects as go
import joblib
```

```
df = pd.read_csv("Kaggle_TED_video_metadata_balanced.csv")
df
```

|     | title \                                            | tags | views   | likes |
|-----|----------------------------------------------------|------|---------|-------|
| 0   | Stories from a home for terminally ill children... |      |         |       |
| 1   | Why our screens make us less happy   Adam Alter    |      |         |       |
| 2   | A tribute to nurses   Carolyn Jones                |      |         |       |
| 3   | Asking for help is a strength, not a weakness ...  |      |         |       |
| 4   | Don't feel sorry for refugees -- believe in th...  |      |         |       |
| ..  | ...                                                |      |         |       |
| 595 | Vincent Moon and Nana Vasconcelos: The world's...  |      |         |       |
| 596 | Maya Beiser(s) and her cello(s)                    |      |         |       |
| 597 | Improving health outcomes with big data   Rond...  |      |         |       |
| 598 | Maybe the best robot demo ever   Marco Tempest     |      |         |       |
| 599 | Niall Ferguson: The 6 killer apps of prosperity    |      |         |       |
|     |                                                    |      |         |       |
|     |                                                    |      |         |       |
| 0   | TED Talk,TED Talks,Children,Community,Death,Fa...  |      | 77455   | 1768  |
| 1   | TEDTalk,TEDTalks,Addiction,Computers,Interface...  |      | 800326  | 20579 |
| 2   | TEDTalk,TEDTalks,Cancer,Community,Compassion,D...  |      | 87635   | 1877  |
| 3   | TEDTalk,TEDTalks,Children,Communication,Commun...  |      | 190840  | 4726  |
| 4   | TEDTalk,TEDTalks,Children,Global issues,Humani...  |      | 98523   | 2669  |
| ..  | ...                                                |      | ...     | ...   |
| 595 | TEDTalk,TEDTalks,TED Talk,TED Talks,TED,TEDGlo...  |      | 82297   | 929   |
| 596 | TED,TEDTalks,Maya Beiser,cello,etude,Steve,Rei...  |      | 28640   | 335   |
| 597 | TEDxTalks,English,Health,Big Data,Data,Illness...  |      | 30288   | 189   |
| 598 | TEDTalk,TEDTalks,TED Talk,TED Talks,TED,TED (0...  |      | 2170703 | 35574 |

```
599 TED,TEDTalks,TEDGlobal,Niall Ferguson,Business... 340865 3757
```

| category_id | dislikes | comment_count | published_at         | duration |
|-------------|----------|---------------|----------------------|----------|
| 0           | 0        | 49            | 2017-03-24T15:32:48Z | PT15M19S |
| 29          |          |               |                      |          |
| 1           | 0        | 572           | 2017-08-01T15:29:04Z | PT9M30S  |
| 22          |          |               |                      |          |
| 2           | 0        | 48            | 2017-05-30T18:17:56Z | PT10M49S |
| 22          |          |               |                      |          |
| 3           | 0        | 187           | 2017-04-12T15:17:51Z | PT11M56S |
| 22          |          |               |                      |          |
| 4           | 0        | 226           | 2017-07-25T15:06:25Z | PT14M14S |
| 29          |          |               |                      |          |
| ..          | ...      | ...           | ...                  | ...      |
| ...         |          |               |                      |          |
| 595         | 0        | 71            | 2014-11-14T16:48:14Z | PT24M18S |
| 10          |          |               |                      |          |
| 596         | 0        | 61            | 2011-06-17T21:14:09Z | PT20M30S |
| 28          |          |               |                      |          |
| 597         | 0        | 4             | 2018-12-05T16:03:49Z | PT10M34S |
| 29          |          |               |                      |          |
| 598         | 0        | 1817          | 2014-05-06T15:35:02Z | PT6M19S  |
| 28          |          |               |                      |          |
| 599         | 0        | 526           | 2011-09-19T16:47:08Z | PT20M20S |
| 25          |          |               |                      |          |

```
[600 rows x 9 columns]
```

## Data voorbereiden

Dislikes zijn niet relevant in deze analyse. Deze data is namelijk niet beschikbaar van Youtube videos. Daarom verwijder ik ze uit de dataset.

```
df.pop("dislikes")
```

```
0      0
1      0
2      0
3      0
4      0
..
595    0
596    0
597    0
598    0
599    0
Name: dislikes, Length: 600, dtype: int64
```

Eventuele duplicaten verwijderen gebaseerd op titel en uploaddatum:

```
df.drop_duplicates(subset=['title', 'published_at'])
```

|     | title \                                           | tags                 | views    | likes       |
|-----|---------------------------------------------------|----------------------|----------|-------------|
| 0   | Stories from a home for terminally ill childre... |                      |          |             |
| 1   | Why our screens make us less happy   Adam Alter   |                      |          |             |
| 2   | A tribute to nurses   Carolyn Jones               |                      |          |             |
| 3   | Asking for help is a strength, not a weakness ... |                      |          |             |
| 4   | Don't feel sorry for refugees -- believe in th... |                      |          |             |
| ..  | ...                                               |                      |          |             |
| 595 | Vincent Moon and Nana Vasconcelos: The world's... |                      |          |             |
| 596 | Maya Beiser(s) and her cello(s)                   |                      |          |             |
| 597 | Improving health outcomes with big data   Rond... |                      |          |             |
| 598 | Maybe the best robot demo ever   Marco Tempest    |                      |          |             |
| 599 | Niall Ferguson: The 6 killer apps of prosperity   |                      |          |             |
|     |                                                   | tags                 | views    | likes       |
| \   |                                                   |                      |          |             |
| 0   | TED Talk,TED Talks,Children,Community,Death,Fa... |                      | 77455    | 1768        |
| 1   | TEDTalk,TEDTalks,Addiction,Computers,Interface... |                      | 800326   | 20579       |
| 2   | TEDTalk,TEDTalks,Cancer,Community,Compassion,D... |                      | 87635    | 1877        |
| 3   | TEDTalk,TEDTalks,Children,Communication,Commun... |                      | 190840   | 4726        |
| 4   | TEDTalk,TEDTalks,Children,Global issues,Humani... |                      | 98523    | 2669        |
| ..  | ...                                               |                      | ...      | ...         |
| 595 | TEDTalk,TEDTalks,TED Talk,TED Talks,TED,TEDGlo... |                      | 82297    | 929         |
| 596 | TED,TEDTalks,Maya Beiser,cello,etude,Steve,Rei... |                      | 28640    | 335         |
| 597 | TEDxTalks,English,Health,Big Data,Data,Illness... |                      | 30288    | 189         |
| 598 | TEDTalk,TEDTalks,TED Talk,TED Talks,TED,TED (0... |                      | 2170703  | 35574       |
| 599 | TED,TEDTalks,TEDGlobal,Niall Ferguson,Business... |                      | 340865   | 3757        |
|     | comment_count                                     | published_at         | duration | category_id |
| 0   | 49                                                | 2017-03-24T15:32:48Z | PT15M19S | 29          |
| 1   | 572                                               | 2017-08-01T15:29:04Z | PT9M30S  | 22          |
| 2   | 48                                                | 2017-05-30T18:17:56Z | PT10M49S | 22          |
| 3   | 187                                               | 2017-04-12T15:17:51Z | PT11M56S | 22          |
| 4   | 226                                               | 2017-07-25T15:06:25Z | PT14M14S | 29          |
| ..  | ...                                               | ...                  | ...      | ...         |
| 595 | 71                                                | 2014-11-14T16:48:14Z | PT24M18S | 10          |
| 596 | 61                                                | 2011-06-17T21:14:09Z | PT20M30S | 28          |

|     |      |                      |          |    |
|-----|------|----------------------|----------|----|
| 597 | 4    | 2018-12-05T16:03:49Z | PT10M34S | 29 |
| 598 | 1817 | 2014-05-06T15:35:02Z | PT6M19S  | 28 |
| 599 | 526  | 2011-09-19T16:47:08Z | PT20M20S | 25 |

[598 rows x 8 columns]

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 600 entries, 0 to 599
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   title                 600 non-null   object
1   tags                  590 non-null   object
2   views                 600 non-null   int64
3   likes                 600 non-null   int64
4   comment_count         600 non-null   int64
5   published_at          600 non-null   object
6   duration              600 non-null   object
7   category_id           600 non-null   int64
dtypes: int64(4), object(4)
memory usage: 37.6+ KB
```

Soorten waarden:

- Views, likes en comment\_count zijn numerieke waarden, dus die hoeven niet aangepast te worden.
- Tags en category\_id zijn categorische waarden die op de one-hot manier zullen ge-encodeerd worden
- Duration zal meteen in de schaler naar numerieke waarden omgezet worden.
- Published\_at zal een tijd encoding krijgen

```
missende_waarden = df.isin([np.nan, None, 'NA', 'null', '']).sum()
print(missende_waarden)
```

```
title          0
tags           10
views          0
likes          0
comment_count  0
published_at   0
duration       0
category_id    0
dtype: int64
```

## Tags

'Tags' heeft missende waarden. Deze zal ik vullen met no\_tag:

```

df['tags'] = df['tags'].fillna('no_tag')
df['tags'] = df['tags'].str.split(',')
df_exploded = df.explode('tags')

one_hot = pd.get_dummies(df_exploded['tags'], prefix='tag')
df_encoded = df_exploded.drop('tags',
axis=1).join(one_hot).groupby(level=0).max()
print(df_encoded.head())
df = df_encoded

```

|   | title                                             | views  | likes | \ |
|---|---------------------------------------------------|--------|-------|---|
| 0 | Stories from a home for terminally ill childre... | 77455  | 1768  |   |
| 1 | Why our screens make us less happy   Adam Alter   | 800326 | 20579 |   |
| 2 | A tribute to nurses   Carolyn Jones               | 87635  | 1877  |   |
| 3 | Asking for help is a strength, not a weakness ... | 190840 | 4726  |   |
| 4 | Don't feel sorry for refugees -- believe in th... | 98523  | 2669  |   |

|                  | comment_count | published_at         | duration | category_id |
|------------------|---------------|----------------------|----------|-------------|
| tag_#NotGuilty \ |               |                      |          |             |
| 0                | 49            | 2017-03-24T15:32:48Z | PT15M19S | 29          |
| False            |               |                      |          |             |
| 1                | 572           | 2017-08-01T15:29:04Z | PT9M30S  | 22          |
| False            |               |                      |          |             |
| 2                | 48            | 2017-05-30T18:17:56Z | PT10M49S | 22          |
| False            |               |                      |          |             |
| 3                | 187           | 2017-04-12T15:17:51Z | PT11M56S | 22          |
| False            |               |                      |          |             |
| 4                | 226           | 2017-07-25T15:06:25Z | PT14M14S | 29          |
| False            |               |                      |          |             |

|                   | tag_#NotGuilty | campaign | tag_'Passing | ...   | tag_испанский |
|-------------------|----------------|----------|--------------|-------|---------------|
| tag_итальянский \ |                |          |              |       |               |
| 0                 | False          | False    | ...          | False |               |
| False             |                |          |              |       |               |
| 1                 | False          | False    | ...          | False |               |
| False             |                |          |              |       |               |
| 2                 | False          | False    | ...          | False |               |
| False             |                |          |              |       |               |
| 3                 | False          | False    | ...          | False |               |
| False             |                |          |              |       |               |
| 4                 | False          | False    | ...          | False |               |
| False             |                |          |              |       |               |

|   | tag_китайский | tag_курсы иностранных языков | tag_курсы харьков | \ |
|---|---------------|------------------------------|-------------------|---|
| 0 | False         | False                        | False             |   |
| 1 | False         | False                        | False             |   |
| 2 | False         | False                        | False             |   |
| 3 | False         | False                        | False             |   |
| 4 | False         | False                        | False             |   |

|   | tag_немецкий | tag_французский | tag_японский | tag_ラウルミドン | tag_ラウ |
|---|--------------|-----------------|--------------|------------|--------|
| 0 | False        | False           | False        | False      | False  |
| 1 | False        | False           | False        | False      | False  |
| 2 | False        | False           | False        | False      | False  |
| 3 | False        | False           | False        | False      | False  |
| 4 | False        | False           | False        | False      | False  |

[5 rows x 3114 columns]

### Categorieën

```
df = pd.get_dummies(df, columns=['category_id'], drop_first=True)
```

### Tijd-encoding

```
df['published_at'] = pd.to_datetime(df['published_at'])
```

### Duration

```
label_encoder = LabelEncoder()
df['duration_encoded'] = label_encoder.fit_transform(df['duration'])
df = df.drop('duration', axis=1)
```

### Data schalen

```
scaler = StandardScaler()

features = df[['views', 'likes', 'comment_count']]
scaled_features = scaler.fit_transform(features)
```

## Eerste analyse

```
kmeans = KMeans(n_clusters=2, random_state=42)
df['Populariteit'] = kmeans.fit_predict(scaled_features)

plt.figure(figsize=(15, 5))

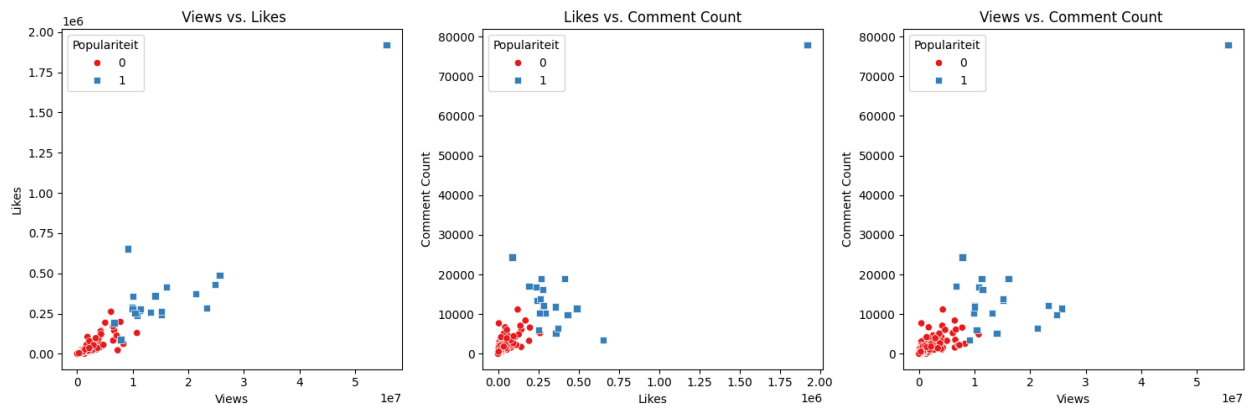
# Scatterplot voor Likes vs Views
plt.subplot(1, 3, 1)
sns.scatterplot(x='views', y='likes', hue='Populariteit', data=df,
palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Views vs. Likes')
plt.xlabel('Views')
plt.ylabel('Likes')

# Scatterplot voor Likes vs Comment Count
```

```
plt.subplot(1, 3, 2)
sns.scatterplot(x='likes', y='comment_count', hue='Populariteit',
data=df, palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Likes vs. Comment Count')
plt.xlabel('Likes')
plt.ylabel('Comment Count')

# Scatterplot voor Comment_count vs Views
plt.subplot(1, 3, 3)
sns.scatterplot(x='views', y='comment_count', hue='Populariteit',
data=df, palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Views vs. Comment Count ')
plt.xlabel('Views')
plt.ylabel('Comment Count')

plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(15, 5))

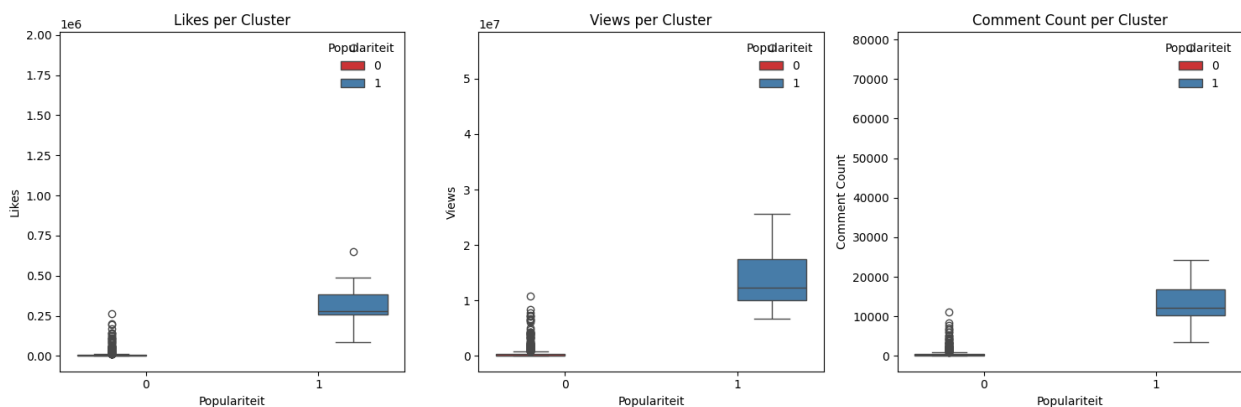
# Boxplot voor Likes
plt.subplot(1, 3, 1)
sns.boxplot(x='Populariteit', y='likes', data=df, hue='Populariteit',
palette='Set1', dodge=True)
plt.title('Likes per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Likes')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

# Boxplot voor Views
plt.subplot(1, 3, 2)
sns.boxplot(x='Populariteit', y='views', data=df, hue='Populariteit',
palette='Set1', dodge=True)
plt.title('Views per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Views')
```

```
plt.legend(title='Populariteit', loc='upper right', frameon=False)

# Boxplot voor Comment Count
plt.subplot(1, 3, 3)
sns.boxplot(x='Populariteit', y='comment_count', data=df,
hue='Populariteit', palette='Set1', dodge=True)
plt.title('Comment Count per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Comment Count')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

plt.tight_layout()
plt.show()
```



Er zit een hele duidelijke uitschieter in de data, die hoogstwaarschijnlijk ook een invloed heeft op de K-means clustering. Er is een video die ver boven het derde kwartiel zit, met 55.62~ views.

## Uitschieter(s)

Om de uitschieters te onderzoeken, is er een extra onderzoek nodig. Het lijkt erop dat *views* en *comment\_count* allebei uitschieters hebben.

```
df[['likes', 'views', 'comment_count']].describe()
```

|       | likes        | views        | comment_count |
|-------|--------------|--------------|---------------|
| count | 6.000000e+02 | 6.000000e+02 | 600.000000    |
| mean  | 2.263473e+04 | 1.066151e+06 | 1070.870000   |
| std   | 1.000825e+05 | 3.611008e+06 | 4055.503954   |
| min   | 0.000000e+00 | 1.000000e+00 | 0.000000      |
| 25%   | 6.242500e+02 | 5.922150e+04 | 86.000000     |
| 50%   | 1.846000e+03 | 1.332595e+05 | 202.000000    |
| 75%   | 6.382250e+03 | 4.101668e+05 | 511.750000    |
| max   | 1.921445e+06 | 5.562224e+07 | 77980.000000  |

Beginnende met *views*:



```

Q1 = df['views'].quantile(0.25)
Q3 = df['views'].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

df_cleaned_views = df[(df['views'] >= lower_bound) & (df['views'] <=
upper_bound)]

```

*Comment\_count:*

```

Q1 = df_cleaned_views['comment_count'].quantile(0.25)
Q3 = df_cleaned_views['comment_count'].quantile(0.75)
IQR = Q3 - Q1

df_cleaned_comments =
df_cleaned_views[(df_cleaned_views['comment_count'] >= lower_bound) &
(df_cleaned_views['comment_count'] <= upper_bound)]
df = df_cleaned_comments

```

## Tweede analyse

De features moeten opnieuw ge-scaled worden en de k-means clustering moet ook opnieuw.

```

features = df[['views', 'likes', 'comment_count']]

scaled_features = scaler.fit_transform(features)
kmeans = KMeans(n_clusters=2, random_state=42)
df['Populariteit'] = kmeans.fit_predict(scaled_features)

plt.figure(figsize=(15, 5))

# Scatterplot voor Likes vs Views
plt.subplot(1, 3, 1)
sns.scatterplot(x='views', y='likes', hue='Populariteit', data=df,
palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Views vs. Likes')
plt.xlabel('Views')
plt.ylabel('Likes')

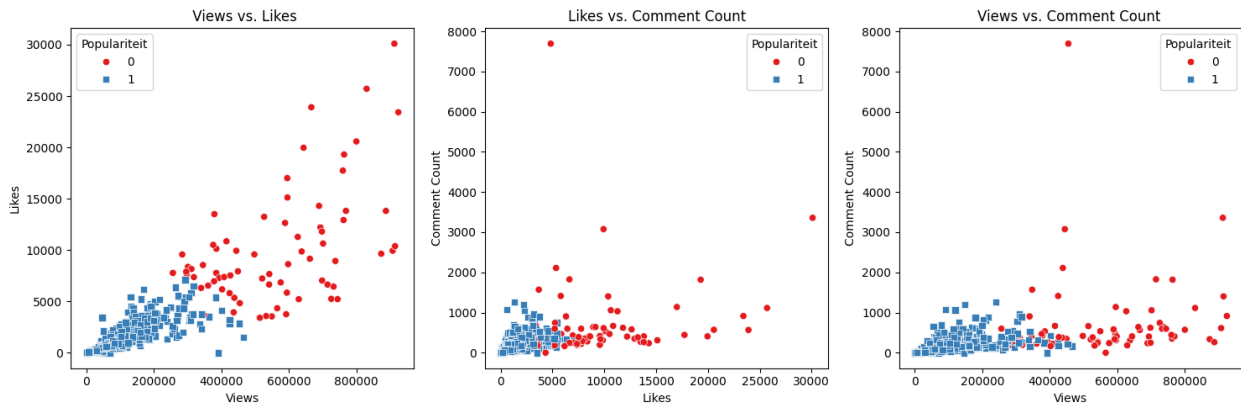
# Scatterplot voor Likes vs Comment Count
plt.subplot(1, 3, 2)
sns.scatterplot(x='likes', y='comment_count', hue='Populariteit',
data=df, palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Likes vs. Comment Count')
plt.xlabel('Likes')
plt.ylabel('Comment Count')

# Scatterplot voor Comment_count vs Views

```

```
plt.subplot(1, 3, 3)
sns.scatterplot(x='views', y='comment_count', hue='Populariteit',
data=df, palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Views vs. Comment Count ')
plt.xlabel('Views')
plt.ylabel('Comment Count')

plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(15, 5))

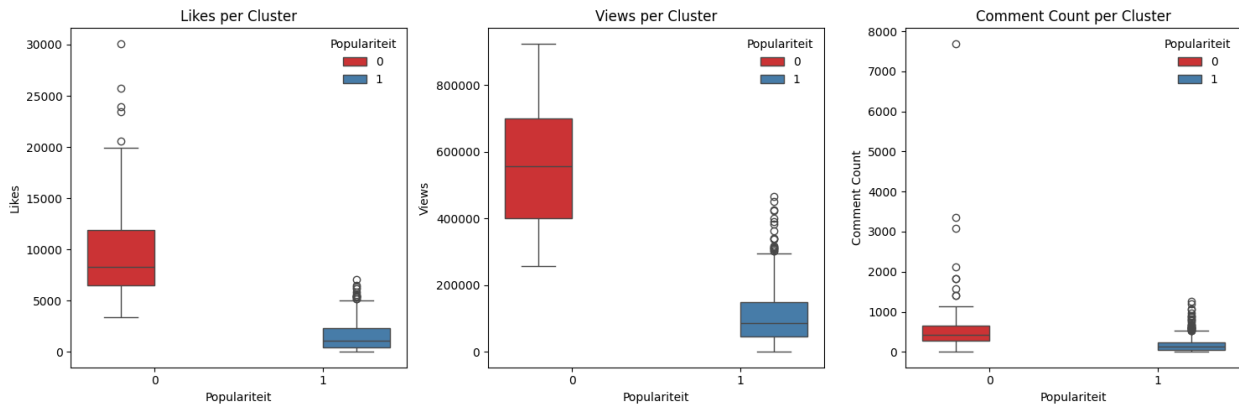
# Boxplot voor Likes
plt.subplot(1, 3, 1)
sns.boxplot(x='Populariteit', y='likes', data=df, hue='Populariteit',
palette='Set1', dodge=True)
plt.title('Likes per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Likes')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

# Boxplot voor Views
plt.subplot(1, 3, 2)
sns.boxplot(x='Populariteit', y='views', data=df, hue='Populariteit',
palette='Set1', dodge=True)
plt.title('Views per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Views')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

# Boxplot voor Comment Count
plt.subplot(1, 3, 3)
sns.boxplot(x='Populariteit', y='comment_count', data=df,
hue='Populariteit', palette='Set1', dodge=True)
plt.title('Comment Count per Cluster')
plt.xlabel('Populariteit')
```

```
plt.ylabel('Comment Count')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

plt.tight_layout()
plt.show()
```



```
scaled_features_df = pd.DataFrame(scaled_features,
columns=['views_scaled', 'likes_scaled', 'comment_count_scaled',
])
correlation_matrix = scaled_features_df.corr()
correlation_matrix.reset_index(inplace=True)
print(correlation_matrix)
```

|                      | index                | views_scaled | likes_scaled |
|----------------------|----------------------|--------------|--------------|
| comment_count_scaled |                      |              |              |
| 0                    | views_scaled         | 1.000000     | 0.851338     |
|                      |                      |              | 0.443567     |
| 1                    | likes_scaled         | 0.851338     | 1.000000     |
|                      |                      |              | 0.422239     |
| 2                    | comment_count_scaled | 0.443567     | 0.422239     |
|                      |                      |              | 1.000000     |

Hoe dichterbij de 1 of -1 staat, hoe sterker de correlatie is. Het lijkt erop dat er een sterke positieve correlatie is tussen likes en views. Comments, views en likes hebben verder niet een sterke correlatie want  $r$  ligt < 0.5. Het zou dus een goed idee zijn om een extra feature te creëren gebaseerd op de likes per view bijvoorbeeld.

Ik wil nu kijken of *duration* ook belangrijk is om toe te voegen:

```
features = df[['views', 'likes', 'comment_count',
'duration_encoded']]

scaled_features = scaler.fit_transform(features)
```

```

kmeans = KMeans(n_clusters=2, random_state=42)
df['Populariteit'] = kmeans.fit_predict(scaled_features)

scaled_features_df = pd.DataFrame(scaled_features,
columns=['views_scaled', 'likes_scaled', 'comment_count_scaled',
'duration_scaled'
])

correlation_matrix = scaled_features_df.corr()

correlation_matrix.reset_index(inplace=True)

print(correlation_matrix)

```

|                        | index                | views_scaled | likes_scaled |
|------------------------|----------------------|--------------|--------------|
| comment_count_scaled \ |                      |              |              |
| 0                      | views_scaled         | 1.000000     | 0.851338     |
| 0.443567               |                      |              |              |
| 1                      | likes_scaled         | 0.851338     | 1.000000     |
| 0.422239               |                      |              |              |
| 2                      | comment_count_scaled | 0.443567     | 0.422239     |
| 1.000000               |                      |              |              |
| 3                      | duration_scaled      | -0.063023    | -0.114484    |
| 0.051977               |                      |              |              |

|   | duration_scaled |
|---|-----------------|
| 0 | -0.063023       |
| 1 | -0.114484       |
| 2 | -0.051977       |
| 3 | 1.000000        |

```

plt.figure(figsize=(15, 5))

# Scatterplot voor Likes vs Views
plt.subplot(1, 3, 1)
sns.scatterplot(x='duration_encoded', y='likes', hue='Populariteit',
data=df, palette='Set1', style='Populariteit', markers=['o', 's'])
plt.title('Views vs. Likes')
plt.xlabel('Duration')
plt.ylabel('Likes')

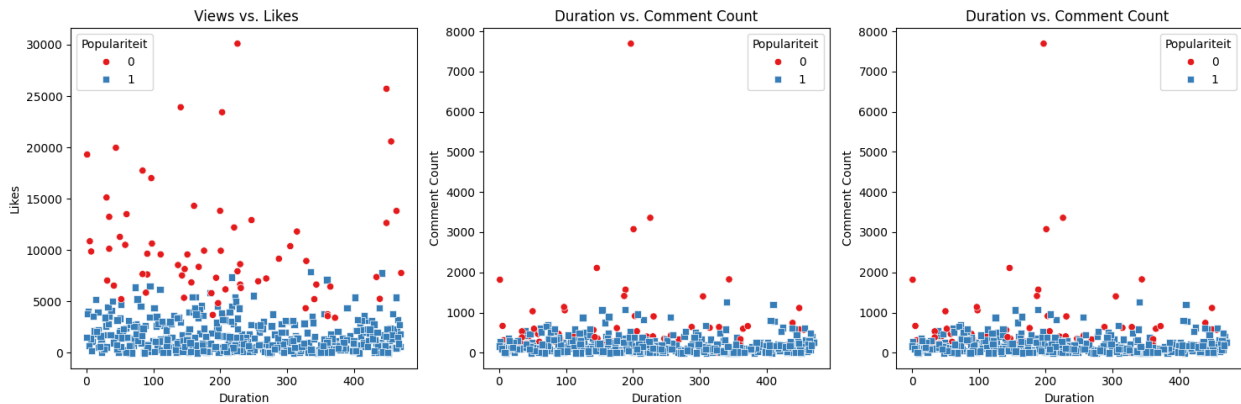
# Scatterplot voor Likes vs Comment Count
plt.subplot(1, 3, 2)
sns.scatterplot(x='duration_encoded', y='comment_count',
hue='Populariteit', data=df, palette='Set1', style='Populariteit',
markers=['o', 's'])
plt.title('Duration vs. Comment Count')
plt.xlabel('Duration')
plt.ylabel('Comment Count')

# Scatterplot voor Comment_count vs Views
plt.subplot(1, 3, 3)

```

```
sns.scatterplot(x='duration_encoded', y='comment_count',
hue='Populariteit', data=df, palette='Set1', style='Populariteit',
markers=['o', 's'])
plt.title('Duration vs. Comment Count ')
plt.xlabel('Duration')
plt.ylabel('Comment Count')

plt.tight_layout()
plt.show()
```



Duration lijkt geen (sterke) associatie te hebben met de drie variabelen. Het heeft dus geen (grote) invloed op of iets populair is of niet. Je kunt wel zien in de correlatie-tabel dat er een zwakke negatieve associatie is met de andere variabelen.

### Nieuwe feature

Uit onderzoek is gebleken dat er een sterke positieve associatie is tussen views en likes. Dit betekent dat het wellicht een goed idee is om een feature toe te voegen genaamd *like\_view\_ratio*.

```
df['like_view_ratio'] = df['likes'] / df['views']
print(df[['likes', 'views', 'like_view_ratio']].describe())
```

|       | likes        | views         | like_view_ratio |
|-------|--------------|---------------|-----------------|
| count | 506.000000   | 506.000000    | 506.000000      |
| mean  | 2715.209486  | 172064.820158 | 0.013969        |
| std   | 3843.658693  | 190792.533590 | 0.007741        |
| min   | 0.000000     | 1.000000      | 0.000000        |
| 25%   | 515.000000   | 51839.250000  | 0.008627        |
| 50%   | 1386.500000  | 104969.000000 | 0.012652        |
| 75%   | 3198.000000  | 194191.250000 | 0.018082        |
| max   | 30102.000000 | 924776.000000 | 0.074279        |

```
features = df[['views', 'likes', 'comment_count', 'like_view_ratio']]
scaled_features = scaler.fit_transform(features)
```

```

kmeans = KMeans(n_clusters=2, random_state=42)
df['Populariteit'] = kmeans.fit_predict(scaled_features)

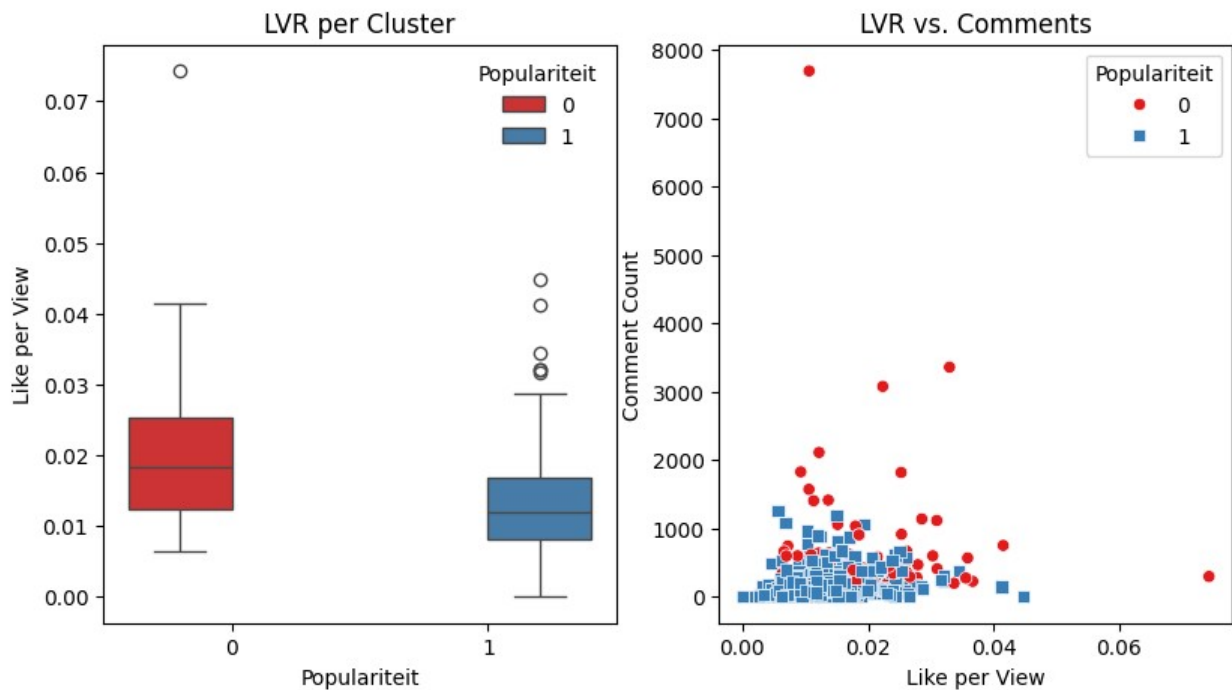
plt.figure(figsize=(15, 5))

plt.subplot(1, 3, 1)
sns.boxplot(x='Populariteit', y='like_view_ratio', data=df,
hue='Populariteit', palette='Set1', dodge=True)
plt.title('LVR per Cluster')
plt.xlabel('Populariteit')
plt.ylabel('Like per View')
plt.legend(title='Populariteit', loc='upper right', frameon=False)

plt.subplot(1, 3, 2)
sns.scatterplot(x='like_view_ratio', y='comment_count',
hue='Populariteit', data=df, palette='Set1', style='Populariteit',
markers=['o', 's'])
plt.title('LVR vs. Comments')
plt.xlabel('Like per View')
plt.ylabel('Comment Count')

Text(0, 0.5, 'Comment Count')

```



```

scaled_features_df = pd.DataFrame(scaled_features,
columns=['views_scaled', 'likes_scaled', 'comment_count_scaled',
'like_view_ratio'
])

correlation_matrix = scaled_features_df.corr()

```

```
correlation_matrix.reset_index(inplace=True)
```

```
print(correlation_matrix)
```

```
          index  views_scaled  likes_scaled
comment_count_scaled \
0          views_scaled      1.000000      0.851338
0.443567
1          likes_scaled      0.851338      1.000000
0.422239
2  comment_count_scaled      0.443567      0.422239
1.000000
3      like_view_ratio      0.211431      0.499598
0.156931
```

```
      like_view_ratio
0      0.211431
1      0.499598
2      0.156931
3      1.000000
```

```
df[['views', 'likes', 'comment_count',
'duration_encoded', 'like_view_ratio']].describe()
```

```
          views      likes  comment_count  duration_encoded \
count      506.000000    506.000000      506.000000      506.000000
mean    172064.820158  2715.209486    263.254941    230.092885
std    190792.533590  3843.658693    468.589753    135.431387
min         1.000000      0.000000      0.000000      0.000000
25%     51839.250000    515.000000     75.250000    118.250000
50%    104969.000000   1386.500000    156.000000    217.500000
75%    194191.250000   3198.000000    304.750000    348.750000
max    924776.000000  30102.000000   7694.000000   471.000000
```

```
      like_view_ratio
count      506.000000
mean         0.013969
std         0.007741
min         0.000000
25%         0.008627
50%         0.012652
75%         0.018082
max         0.074279
```

```
# Maak histograms
```

```
plt.figure(figsize=(15, 5))
```

```
# Histogram voor Likes
```

```
plt.subplot(1, 3, 1)
```

```
sns.histplot(df['likes'], bins=10, kde=True) # Voeg een KDE toe voor
```

```

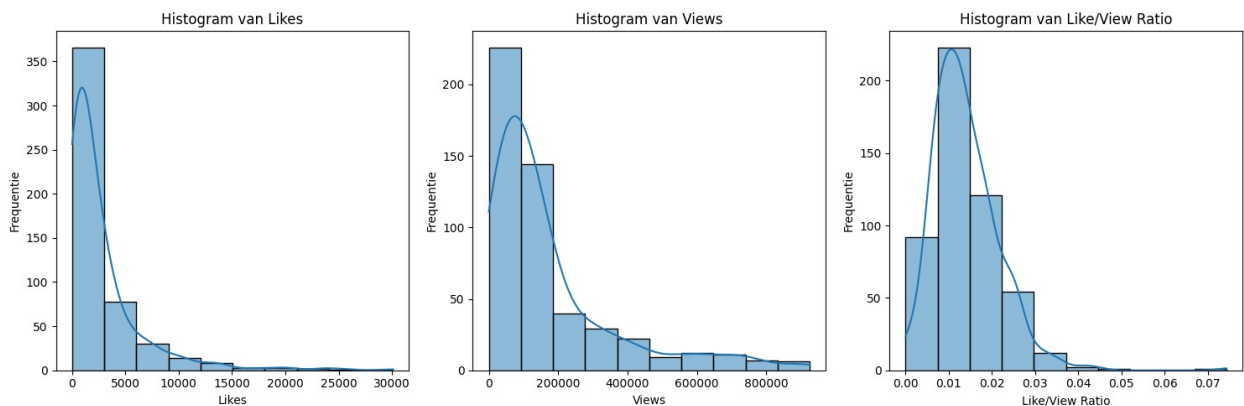
een gladde lijn
plt.title('Histogram van Likes')
plt.xlabel('Likes')
plt.ylabel('Frequentie')

# Histogram voor Views
plt.subplot(1, 3, 2)
sns.histplot(df['views'], bins=10, kde=True)
plt.title('Histogram van Views')
plt.xlabel('Views')
plt.ylabel('Frequentie')

# Histogram voor Like to View Ratio
plt.subplot(1, 3, 3)
sns.histplot(df['like_view_ratio'], bins=10, kde=True)
plt.title('Histogram van Like/View Ratio')
plt.xlabel('Like/View Ratio')
plt.ylabel('Frequentie')

plt.tight_layout()
plt.show()

```



Een meerderheid van de video's ligt onder het gemiddelde, ondanks dat de uitschieters eruit zijn gehaald. Daarom wordt er ook niet naar het gemiddelde gekeken, maar naar de mediaan.

### Schoongemaakte data opslaan

```
df.to_csv('TEDx_schone_data.csv', index=False)
```

## Conclusie

De belangrijkste variabelen die het meest van invloed zijn op de populariteit van video's zijn:

- Views
- Comment\_count
- Likes
- Like\_view\_ratio



Gebaseerd hierop kun je kijken naar of een video populair is of niet.